

Supplementary Material for “Interaction Dynamics: A Physical-Interaction Reasoning Layer for Humanoid Robots”

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I. CONTROLLER EQUATIONS AND PARAMETERS

The simulation runs at 500 Hz and the layer at 50 Hz ($\Delta t = 0.02$ s). The planar-CoM error is $e = c_{xy} - c_{xy}^{\text{ref}}$ with a deadband-shrunk error $e_{\text{eff}} = e \max(0, \|e\| - \delta) / \|e\|$. The external force $\hat{F}_{\text{ext}}$ is estimated by the horizontal wrench estimator of the main paper, low-passed at 5 Hz.

Confidence gate. During quiescence ($\|e\| < \delta$) an EWMA tracks the wrench-noise power, $\sigma_f^2 \leftarrow (1 - \alpha)\sigma_f^2 + \alpha\|\hat{F}_{\text{ext}}\|^2$ with $\alpha = 1 - e^{-2\pi f_\sigma \Delta t}$, and $F_{\text{cap}} = F_{\text{floor}} + k_\sigma \sqrt{\sigma_f^2}$. The latch is

$$g \leftarrow \begin{cases} 1, & \|\hat{F}_{\text{ext}}\| > F_{\text{cap}}, \\ \max(0, g - \Delta t / \tau_g), & \|\hat{F}_{\text{ext}}\| \leq F_{\text{cap}} \wedge \|e\| < 1.5\delta, \\ g, & \text{otherwise,} \end{cases} \quad (1)$$

so capture stays engaged through a long recovery ($\|e\| \geq 1.5\delta$) and releases only once recovered.

Persistence. A wrench timer, decaying at twice the fill rate,

$$t_F \leftarrow \begin{cases} t_F + \Delta t, & \|\hat{F}_{\text{ext}}\| > F_{\text{th}}, \\ \max(0, t_F - 2\Delta t), & \text{else,} \end{cases} \quad p = \text{clip}\left(\frac{t_F - \tau_0}{\tau_r}, 0, 1\right) \quad (2)$$

Capture and hold. a_e is the first input of a horizon- N MPC on $\ddot{e} = a_e + \hat{d}_{\text{eff}}$ minimizing $\sum_j (q_p \|e_j\|^2 + q_v \|\dot{e}_j\|^2 + r \|a_{e,j}\|^2)$ s.t. $\|a_e\| \leq a_{\text{max}}$; then $u_{\text{cap}} = -k_{\text{map}} a_e$ (so k_{map} , in seconds, maps task acceleration to a velocity-command bias). The hold uses a p -gated leaky integral $e_I \leftarrow 0.98 e_I + p e_{\text{eff}} \Delta t$, $u_{\text{hold}} = -k_p e_{\text{eff}} - k_i e_I$, and the output is

$$u = \text{sat}_{v_{\text{max}}}\left(\text{slew}[(1 - p)g u_{\text{cap}} + p u_{\text{hold}}] \odot s_{\text{axis}}\right), \quad (3)$$

added to the frozen policy’s walk command. Parameters are in Table I.

Floor-corrected sustained drift. For seed i and force F , $D_i(F) = |\bar{o}_i(F) - \bar{o}_i(0)|$, where $\bar{o}_i(F)$ is the settled lateral CoM offset from the recorded reference and $\bar{o}_i(0)$ the same-seed no-force run; we report the median [IQR] over seeds. Same-seed subtraction cancels the gait-sway/reference-phase floor (~ 30 mm).

Why a persistent bias does not fire the gate. A constant foot-force bias b is absorbed by σ_f : during quiescence the EWMA drives $\sigma_f^2 \rightarrow b^2$, so $F_{\text{cap}} \rightarrow F_{\text{floor}} + k_\sigma b \approx 16$ N at $b = 5$ N, far above the bias—hence $\|\hat{F}_{\text{ext}}\| = b$ never crosses F_{cap} and capture is not engaged. Sustained rejection is unaffected because the hold path is deviation-driven and ungated. A slowly-updated bias estimate μ_F with the gate on

TABLE I
LAYER PARAMETERS (AS IMPLEMENTED).

symbol	meaning	value
Δt	control period	0.02 s (50 Hz)
δ	error deadband	0.03 m
F_{floor}	force floor	1.0 N
k_σ	confidence margin	3.0
f_σ	noise-tracker bandwidth	0.3 Hz
τ_g	gate-release window	1.6 s
F_{th}	persistence threshold	3.0 N
τ_0, τ_r	persistence delay, ramp	0.4 s, 0.4 s
N	MPC horizon	20 (0.4 s)
q_p, q_v, r	MPC weights	15, 100, 0.05
a_{max}	MPC accel bound	6 m/s ²
k_{map}	accel→cmd map	0.12 s
k_p, k_i	hold P, I	3.0, 8.0
s_{axis}	(x, y) channel scale	(0.2, 1.0)
$v_{\text{max}}, \text{slew}$	bias clamp, rate limit	0.6 m/s, 6.0 m/s ²

Algorithm 1 Confidence-gated capture–hold arbitration (per 50 Hz step)

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1:  $\hat{F}_{\text{ext}} \leftarrow m\ddot{c}_{xy} - \sum_i F_{i,xy}$   $\triangleright$  estimate; low-pass 5 Hz
2: if  $\|e\| < \delta$  then update  $\sigma_f^2$  (EWMA); return  $u = 0$   $\triangleright$  quiescent
3:  $F_{\text{cap}} \leftarrow F_{\text{floor}} + k_\sigma \sqrt{\sigma_f^2}$ ; update latch  $g$  by (1)
4: update persistence  $p$  by (2)
5:  $a_e \leftarrow \text{MPC}(e_{\text{eff}}, \dot{e}, \hat{d}_{\text{eff}})$ ;  $u_{\text{cap}} \leftarrow -k_{\text{map}} a_e$ 
6:  $u_{\text{hold}} \leftarrow -k_p e_{\text{eff}} - k_i e_I$   $\triangleright e_I$ :  $p$ -gated leaky integral
7:  $u \leftarrow (1 - p)g u_{\text{cap}} + p u_{\text{hold}}$ ; then  $\odot s_{\text{axis}}$ , slew-limit, saturate (3)
8: add  $u$  to the policy’s velocity command

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$\|\hat{F}_{\text{ext}} - \mu_F\|$ would separate bias from force explicitly; we note this as the cleaner design for hardware.